



INTRODUCTION TO DATA STRUCTURE S

MIT SCHOOL OF COMPUTING

AS PART OF CURRICULAM

FIRST YEAR (SEM-2)

COMPUTER SCIENCEAND ENGINEERING

UNDER THE GUIDANCE OF

Prof. Sachin Tiwari

FOR THE ACADEMIC YEAR 2025-26

SUBMITTED BY

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EXPERIMENT DETAILS

Experiment Number	Experiment Name
	mg e e 1st
2	Doubly Linked List
3	

Virtual Lab NameVirtual Labs

:

Institute:

111T-H(Hyderabad)

Date of Performance: 28 M arch 2026

SCREENSHOTS

Computer Science and Engineering > Data Structures > Experiments

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Aim

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Linked List

Choose difficulty: ☒ Beginner ☒ Intermediate

1. Which of these best describes an array?

☐ a. A data structure that shows a hierarchical behaviour [Explanation](#)

☒ b. Container of objects of uniform types [Explanation](#)

☐ c. Container of objects of mixed types [Explanation](#)

☐ d. All of the mentioned [Explanation](#)

2. What is the time complexity of insertion at any point in an array?

☒ a. $O(1)$

☐ b. $O(N^2)$

☐ c. $O(N \log N)$

☐ d. None of these

3. Pick incorrect options from the following

☐ a. If structure contains 3 types of data then one pointer is sufficient for that structure [Explanation](#)

☒ b. One variable can be bound to many structures [Explanation](#)

☐ c. Contents of a structure are stored in contiguous memory [Explanation](#)

☐ d. None of these

4. The memory occupied by a pointer _____

☐ a. Is dependent on which data type the pointer points to [Explanation](#)

☒ b. Is independent of data type [Explanation](#)

☐ c. Depends on the program we are writing [Explanation](#)

☐ d. None of these

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Score: 4 out of 4

MAAF

Linked List

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Instructions

Question Count: 10, 11, 12, 13, 14, 15, 16, 17, 18

h/cad



Linked List

Insert All Nodes

Insert At Tail

Insert At Head

Overview: **Pretest**

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Choose difficulty: ☒ Beginner ☐ Intermediate

1. As we have the memory address of Nodes can we traverse backwards in a linked list?

☐ a Yes we can [Explanation](#)

☒ b No we can't [Explanation](#)

2. What are the advantages of a linked list?

☒ a They require less memory than an array to store the same data [Explanation](#)

☐ b They require less memory than an array to store the same data [Explanation](#)

☐ c We can easily traverse back to previous elements [Explanation](#)

☐ d None of the above [Explanation](#)

3. What is the time complexity for insertion of an element into linked list?

☐ a O(1) [Explanation](#)

☐ b O(log n) [Explanation](#)

☒ c O(n) [Explanation](#)

☐ d O(n^2) [Explanation](#)

4. The following is a pseudo code for deleting an element from a linked list

```

1. //
2. // A pseudo code for deleting an element at the end
3. // head -> pointer pointing to the first node
4. // cur -> a temporary pointer
5. // prev -> a pointer pointing to the node before cur
6. //
7. deleteNode() {
8.     cur = head
9.     while (cur->next != NULL) {
10        prev = cur
11        cur = cur->next
12    }
13    // Your code here :)
14 }

```

Which of the following is the most appropriate substitution for line 13?

☒ a prev->next = NULL [Explanation](#)

☐ b prev->next = NULL [Explanation](#)

☐ c prev->next = head [Explanation](#)

☐ d cur->next = NULL [Explanation](#)

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ds1-rith.vlabs.ac.in/exp/linked-list/doubly-linked-list/dllexercise.html

Linked List

Instructions

Question: Convert to LL: 19, 42, 5, 81

Observations

Correct

[Insert At Head](#) [Insert At Tail](#)

[Insert Before Node](#) [Submit](#) [Reset](#)

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Computer Science and Engineering Data Structures > Experiments

Choose difficulty: ☒ Beginner ☒ Intermediate

1. What are the advantages of a doubly linked list?

☐ a The memory required for a doubly linked list is less than a linked list [Explanation](#)

☒ b Insertion and deletion are easier in a doubly linked list [Explanation](#)

☐ c Inserting an element is faster in a doubly linked list [Explanation](#)

☐ d There is no advantage [Explanation](#)

[Visit All Nodes](#)

2. What is the time complexity for insertion of an element into a doubly linked list?

☐ a O(1) [Explanation](#)

☐ b O(log n) [Explanation](#)

☒ c O(n) [Explanation](#)

☐ d O(n^2) [Explanation](#)

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Linked List

3. Which of the following can make deletion in doubly linked list a constant time operation?

- ☐ a. Keeping the doubly linked list sorted [Explanation](#)
- ☐ b. Having a pointer at both the first and last element [Explanation](#)
- ☒ c. Having a pointer to the node to be deleted [Explanation](#)
- ☐ d. Knowing the position of the node [Explanation](#)

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Score 3 out of 3

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Linked List

★★★★★

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Instructions

Questions: Convert to 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100

[Insert At Head](#)

[Insert At Tail](#)

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Linked List

1. In a circular linked list

☐ a. We can navigate through the linked list in reverse order

☒ b. We can traverse the linked list in forward order

☐ c. Searching is faster than in a singly linked list

☐ d. None of these

2. In a circular linked list, if we insert an element how many pointers do we have to update?

☐ a. 0

☐ b. 1

☒ c. Expansion

☐ d. 3

3. What is the time complexity for deleting an element from circular linked list?

☐ a. $O(n)$ Expansion

☐ b. $O(\log n)$

☒ c. $O(1)$

☐ d. $O(n^2)$

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Score: 3 out of 3

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Choose difficulty:

☒ Beginner

☒ Intermediate

1. Pick one correct statement from the following?

☐ a. In a doubly linked list, each node is linked to one node

☐ b. In a doubly linked list, each node is linked to three nodes

☒ c. In a doubly linked list, each node is linked to two nodes

☐ d. In a doubly linked list, each node is linked to zero nodes

2. What is the time complexity of searching for an element in a circular linked list?

☒ a. $O(n)$ Expansion

☐ b. $O(\log n)$

☐ c. $O(1)$

☐ d. None of the mentioned

3. Assuming memory is not a limitation which of the following linked list is the best for searching

☐ a. Singly Linked List

☒ b. Singly Linked List Expansion

☐ c. Circular Linked List

☐ d. All give same performance

4. Consider the following linked list: 2 → 4 → 7 → 9 → 10 → 13. Their addresses are 2408, 2560, 1950, 5440, 3240, 3000. Head variable is pointing at 4 (the second node). What will be head → next → next → value?

(Node has value and next, value stores the value and next stores the address of the next variable)

☐ a. 7

☐ b. 10 Expansion

☒ c. 13

☐ d. 13

5. The first node of a linked list is at the location 2000. At which memory location will the second node be added to. Assume the nodes contain integers, each integer occupies a memory of 4 bits. And each pointer is 1 bit.

☐ a. 2004

☐ b. 2008

☐ c. 1999

☒ d. Expansion Expansion

Submit Quiz

Score: 5 out of 5

STUDENT DECLARATION

I here by declare that the Virtual Lab experiments included in this document have been performed by me as part of my academic activity. The screenshots submitted are genuine and taken during the actual execution of the experiments.

Student Signature:



Date:26/04/2026